

C3M1: Peer Reviewed Assignment

Outline:

The objectives for this assignment:

- 1. Apply Binomial regression methods to real data.
- 2. Understand how to analyze and interpret binomial regression models.
- 3. Flex our math skills by determining whether certain distributions are members of the exponential family.

General tips:

- 1. Read the questions carefully to understand what is being asked.
- 2. This work will be reviewed by another human, so make sure that you are clear and concise in what your explanations and answers.

```
In [1]: # Load required libraries
library(tidyverse)
library(dplyr)
```

Attaching packages

tidyverse 1.3.0

✓ ggplot2 3.3.0

✓ purrr 0.3.4

✓ tibble 3.2.1

✓ dplyr 1.1.2

✓ tidyr 1.0.2

✓ stringr 1.4.0

✓ readr 1.3.1

✓ forcats 0.5.0

Conflicts

tidyverse_conflicts()

✗ dplyr::filter()

masks stats::filter()

✗ dplyr::lag()

masks stats::lag()

Problem 1: Binomial (Logistic) Regression

The National Institute of Diabetes and Digestive and Kidney Diseases conducted a study of 768 adult female Pima Indians living near Phoenix, AZ. The purpose of the study was to investigate the factors related to diabetes.

Before we analyze these data, we should note that some have raised ethical issues with its collection and popularity in the statistics and data science community. We should think seriously about these concerns. For example, Maya Iskandarani wrote a brief [piece \(https://researchblog.duke.edu/2016/10/24/diabetes-and-privacy-meet-big-data/\)](https://researchblog.duke.edu/2016/10/24/diabetes-and-privacy-meet-big-data/), on consent and privacy concerns raised by this dataset. After you familiarize yourself with the data, we'll then turn to these ethical concerns.

First, we'll use these data to get some practice with GLM and Logistic regression.

```
In [2]: # Load the data
pima = read.csv("pima.txt", sep="\t")
# Here's a description of the data: https://rdrr.io/cran/faraway/man/pima.html
head(pima)
```

A data.frame: 6 × 9

	pregnant	glucose	diastolic	triceps	insulin	bmi	diabetes	age	test
	<int>	<int>	<int>	<int>	<int>	<dbl>	<dbl>	<int>	<int>
1	6	148	72	35	0	33.6	0.627	50	1
2	1	85	66	29	0	26.6	0.351	31	0
3	8	183	64	0	0	23.3	0.672	32	1
4	1	89	66	23	94	28.1	0.167	21	0
5	0	137	40	35	168	43.1	2.288	33	1
6	5	116	74	0	0	25.6	0.201	30	0

```
In [3]: dim(pima)
```

768 · 9

Diabetes survey on Pima Indians Description

The National Institute of Diabetes and Digestive and Kidney Diseases conducted a study on 768 adult female Pima Indians living near Phoenix. Format

The dataset contains the following variables

- pregnant
- Number of times pregnant
- glucose
- Plasma glucose concentration at 2 hours in an oral glucose tolerance test
- diastolic
- Diastolic blood pressure (mm Hg)
- triceps
- Triceps skin fold thickness (mm)
- insulin
- 2-Hour serum insulin (mu U/ml)
- bmi
- Body mass index (weight in kg/(height in metres squared))
- diabetes
- Diabetes pedigree function
- age
- Age (years)
- test
- test whether the patient shows signs of diabetes (coded 0 if negative, 1 if positive)

1. (a) Data Cleaning? What about Data Scrubbing? Data Sterilizing?

This is a real data set, which means that there's likely going to be gaps and missing values in the data. Before doing any modeling, we should inspect the data and clean it if necessary.

Perform simple graphical and numerical summaries of the data. Pay attention for missing or nonsensical values. Can you find any obvious irregularities? If so, take appropriate steps to correct these problems. In the markdown cell, specify what cleaning you did and why you did it.

Finally, split your data into training and test sets. Let the training set contain 80% of the rows and the test set contain the remaining 20%.

In [4]:

Your Code Here

Summary

In [6]:

summary(pima)

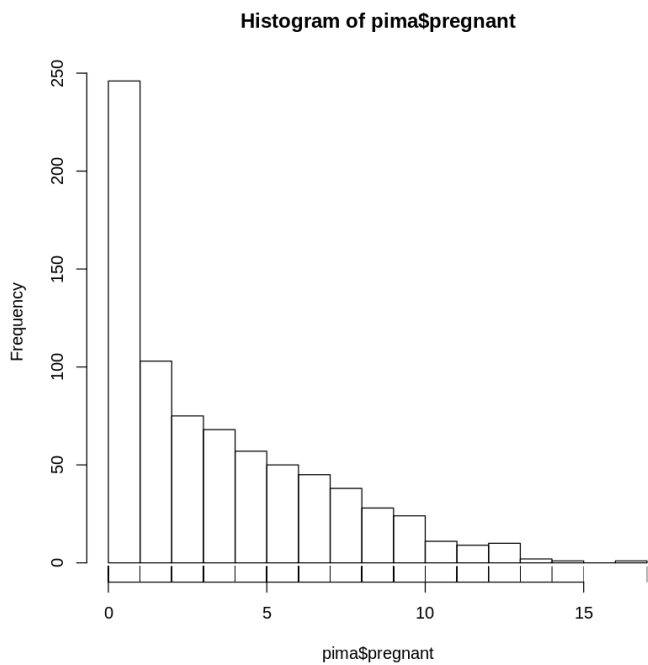
pregnant	glucose	diastolic	triceps
Min. : 0.000	Min. : 0.0	Min. : 0.00	Min. : 0.00
1st Qu.: 1.000	1st Qu.: 99.0	1st Qu.: 62.00	1st Qu.: 0.00
Median : 3.000	Median :117.0	Median : 72.00	Median :23.00
Mean : 3.845	Mean :120.9	Mean : 69.11	Mean :20.54
3rd Qu.: 6.000	3rd Qu.:140.2	3rd Qu.: 80.00	3rd Qu.:32.00
Max. :17.000	Max. :199.0	Max. :122.00	Max. :99.00
insulin	bmi	diabetes	age
Min. : 0.0	Min. : 0.00	Min. :0.0780	Min. :21.00
1st Qu.: 0.0	1st Qu.:27.30	1st Qu.:0.2437	1st Qu.:24.00
Median : 30.5	Median :32.00	Median :0.3725	Median :29.00
Mean : 79.8	Mean :31.99	Mean :0.4719	Mean :33.24
3rd Qu.:127.2	3rd Qu.:36.60	3rd Qu.:0.6262	3rd Qu.:41.00
Max. :846.0	Max. :67.10	Max. :2.4200	Max. :81.00
test			
Min. :0.000			
1st Qu.:0.000			
Median :0.000			
Mean :0.349			
3rd Qu.:1.000			
Max. :1.000			

- So no missing values here.
- But need to explore/understand: pregnant, insulin
- Also, there are many zeros in the wrong places: glucose, diastolic, triceps, bmi

Lets explore these:

Pregnant

```
In [7]: hist(pima$pregnant, breaks = 18)
rug(pima$pregnant)
```



This is total number of pregnancies. Lets look at the age of that woman with 17 pregnancies:

```
In [8]: pima[pima$pregnant>13,]
```

A data.frame: 4 × 9

	pregnant	glucose	diastolic	triceps	insulin	bmi	diabetes	age	test
	<int>	<int>	<int>	<int>	<int>	<dbl>	<dbl>	<int>	<int>
89	15	136	70	32	110	37.1	0.153	43	1
160	17	163	72	41	114	40.9	0.817	47	1
299	14	100	78	25	184	36.6	0.412	46	1
456	14	175	62	30	0	33.6	0.212	38	1

So lets leave this!

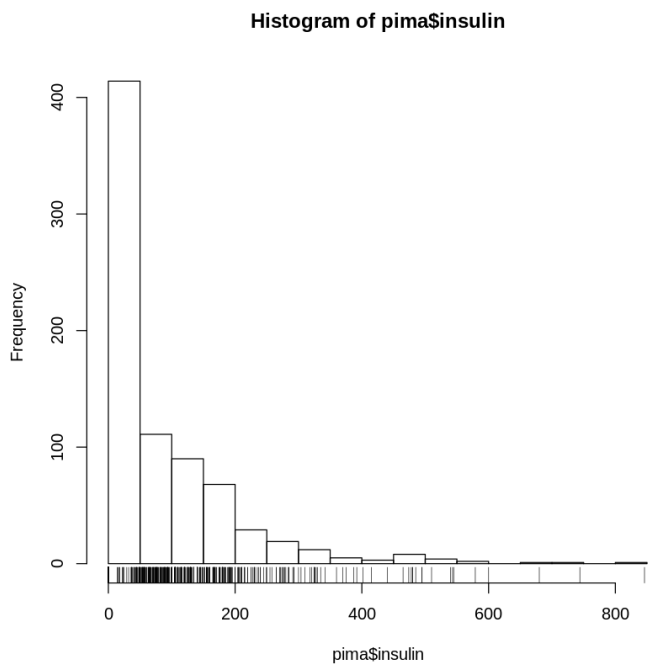
insulin

2-Hour serum insulin (mu U/ml)

Not sure what insulin here means.

- It could mean the level of insulin in their body.
- Or it could mean the insulin being administered.

```
In [9]: hist(pima$insulin, breaks = 18)
rug(pima$insulin)
```



```
In [10]: table(pima$insulin == 0)
```

FALSE	TRUE
394	374

```
In [11]: 100*prop.table(table(pima$insulin == 0))
```

FALSE	TRUE
51.30208	48.69792

- About 49% women are showing zero insulin. So this is most likely insulin being administered.
- Some women seem to be having very high dosage of insulin. Lets look at the age of these:

```
In [11]: pima[pima$insulin>500,]
```

A data.frame: 9 × 9

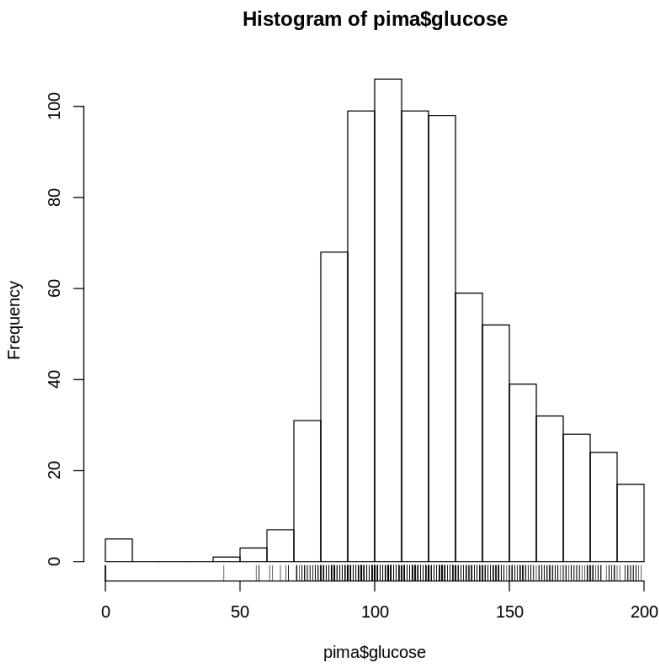
	pregnant	glucose	diastolic	triceps	insulin	bmi	diabetes	age	test
	<int>	<int>	<int>	<int>	<int>	<dbl>	<dbl>	<int>	<int>
9	2	197	70	45	543	30.5	0.158	53	1
14	1	189	60	23	846	30.1	0.398	59	1
229	4	197	70	39	744	36.7	2.329	31	0
248	0	165	90	33	680	52.3	0.427	23	0
287	5	155	84	44	545	38.7	0.619	34	0
410	1	172	68	49	579	42.4	0.702	28	1
585	8	124	76	24	600	28.7	0.687	52	1
656	2	155	52	27	540	38.7	0.240	25	1
754	0	181	88	44	510	43.3	0.222	26	1

It is possible that high dosage of insulin is needed for some people. So lets leave this!

glucose

Plasma glucose concentration at 2 hours in an oral glucose tolerance test.

```
In [12]: hist(pima$glucose, breaks = 18)
rug(pima$glucose)
```



```
In [13]: table(pima$glucose == 0)
```

FALSE	TRUE
763	5

- 5 women are showing zero glucose. This is most likely an error.

```
In [14]: pima[pima$glucose==0,]
```

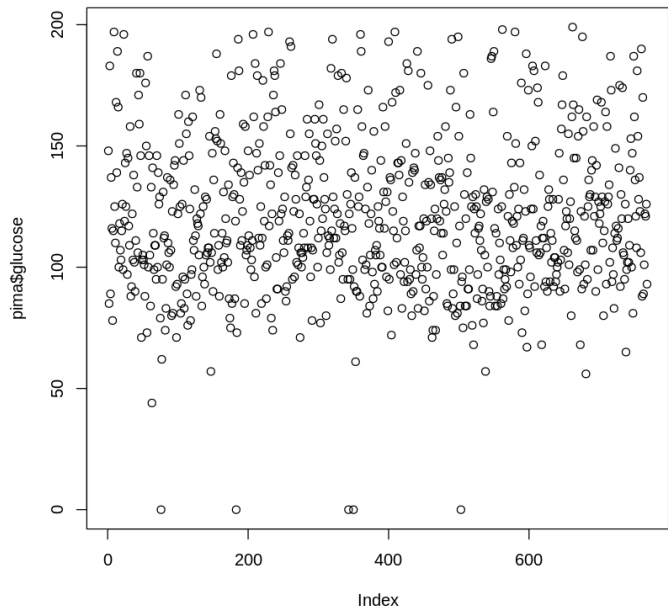
A data.frame: 5 × 9

	pregnant	glucose	diastolic	triceps	insulin	bmi	diabetes	age	test
	<int>	<int>	<int>	<int>	<int>	<dbl>	<dbl>	<int>	<int>
76	1	0	48	20	0	24.7	0.140	22	0
183	1	0	74	20	23	27.7	0.299	21	0
343	1	0	68	35	0	32.0	0.389	22	0
350	5	0	80	32	0	41.0	0.346	37	1
503	6	0	68	41	0	39.0	0.727	41	1

In principle we can drop this. But lets not do that.

Lets see if we can use the data to replace the zero glucose value.

```
In [15]: plot(pima$glucose)
```



Yes. This looks good. Lets replace these zero glucose values with the mean glucose values.

```
In [16]: zeroGlucoseIndices = which(pima$glucose==0)
zeroGlucoseIndices
```

76 · 183 · 343 · 350 · 503

```
In [17]: pima$glucose[zeroGlucoseIndices]
```

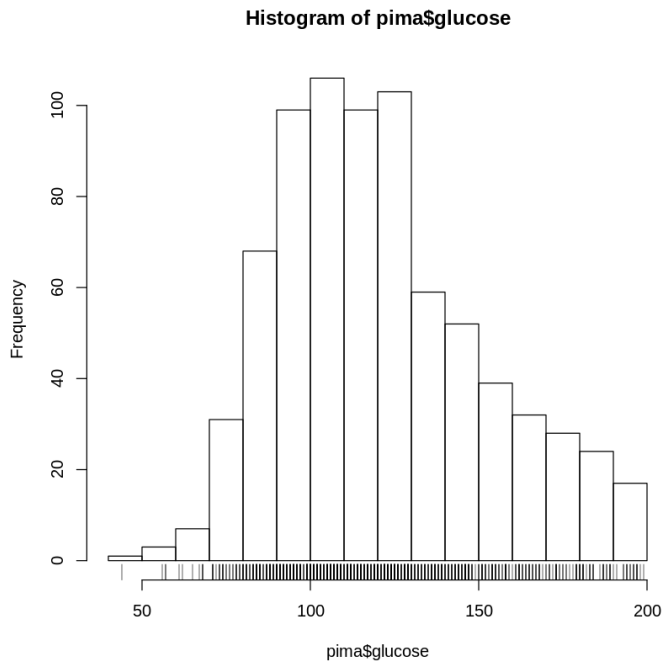
0 · 0 · 0 · 0 · 0

```
In [18]: pima$glucose[zeroGlucoseIndices] = round(mean(pima$glucose[pima$glucose!=0]))
```

```
In [19]: pima$glucose[zeroGlucoseIndices]
```

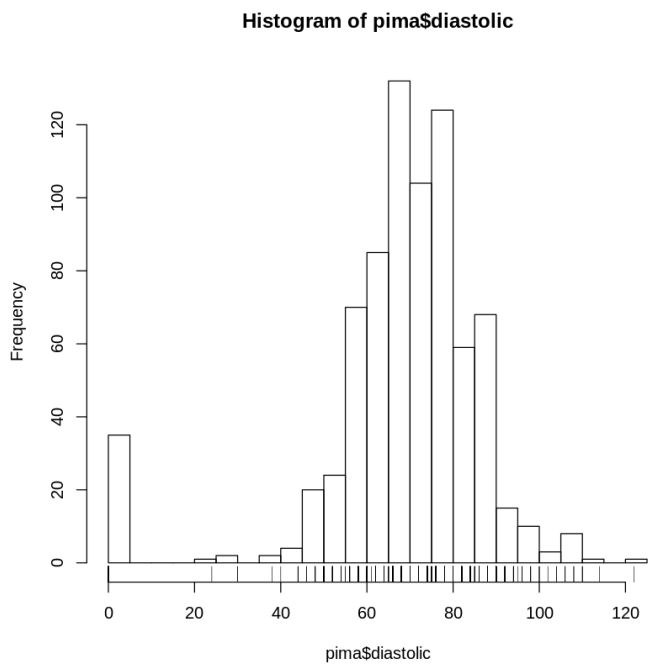
122 · 122 · 122 · 122 · 122

```
In [20]: hist(pima$glucose, breaks = 18)
rug(pima$glucose)
```



diastolic

```
In [21]: hist(pima$diastolic, breaks = 18)
rug(pima$diastolic)
```



```
In [22]: table(pima$diastolic == 0)

FALSE TRUE
 733    35
```

- 35 women are showing zero diastolic. This is most likely an error.

```
In [23]: pima[pima$diastolic==0,]
```

A data.frame: 35 × 9

	pregnant	glucose	diastolic	triceps	insulin	bmi	diabetes	age	test
	<int>	<dbl>	<int>	<int>	<int>	<dbl>	<dbl>	<int>	<int>
8	10	115	0	0	0	35.3	0.134	29	0
16	7	100	0	0	0	30.0	0.484	32	1
50	7	105	0	0	0	0.0	0.305	24	0
61	2	84	0	0	0	0.0	0.304	21	0
79	0	131	0	0	0	43.2	0.270	26	1
82	2	74	0	0	0	0.0	0.102	22	0
173	2	87	0	23	0	28.9	0.773	25	0
194	11	135	0	0	0	52.3	0.578	40	1
223	7	119	0	0	0	25.2	0.209	37	0
262	3	141	0	0	0	30.0	0.761	27	1
267	0	138	0	0	0	36.3	0.933	25	1
270	2	146	0	0	0	27.5	0.240	28	1
301	0	167	0	0	0	32.3	0.839	30	1
333	1	180	0	0	0	43.3	0.282	41	1
337	0	117	0	0	0	33.8	0.932	44	0
348	3	116	0	0	0	23.5	0.187	23	0
358	13	129	0	30	0	39.9	0.569	44	1
427	0	94	0	0	0	0.0	0.256	25	0
431	2	99	0	0	0	22.2	0.108	23	0
436	0	141	0	0	0	42.4	0.205	29	1
454	2	119	0	0	0	19.6	0.832	72	0
469	8	120	0	0	0	30.0	0.183	38	1
485	0	145	0	0	0	44.2	0.630	31	1
495	3	80	0	0	0	0.0	0.174	22	0
523	6	114	0	0	0	0.0	0.189	26	0
534	6	91	0	0	0	29.8	0.501	31	0
536	4	132	0	0	0	32.9	0.302	23	1
590	0	73	0	0	0	21.1	0.342	25	0
602	6	96	0	0	0	23.7	0.190	28	0
605	4	183	0	0	0	28.4	0.212	36	1
620	0	119	0	0	0	32.4	0.141	24	1
644	4	90	0	0	0	28.0	0.610	31	0
698	0	99	0	0	0	25.0	0.253	22	0
704	2	129	0	0	0	38.5	0.304	41	0
707	10	115	0	0	0	0.0	0.261	30	1

In principle we can drop this. But lets not do that.

Lets see if we can use that data to replace the zero diastolic value.

A scatter plot showing the relationship between the index of the data points (x-axis) and the diastolic blood pressure (y-axis, labeled 'pima\$diastolic'). The x-axis ranges from 0 to 700, and the y-axis ranges from 0 to 120. The plot displays a large number of data points, many of which are outliers, particularly at the extremes of the y-axis (near 0 and near 120).

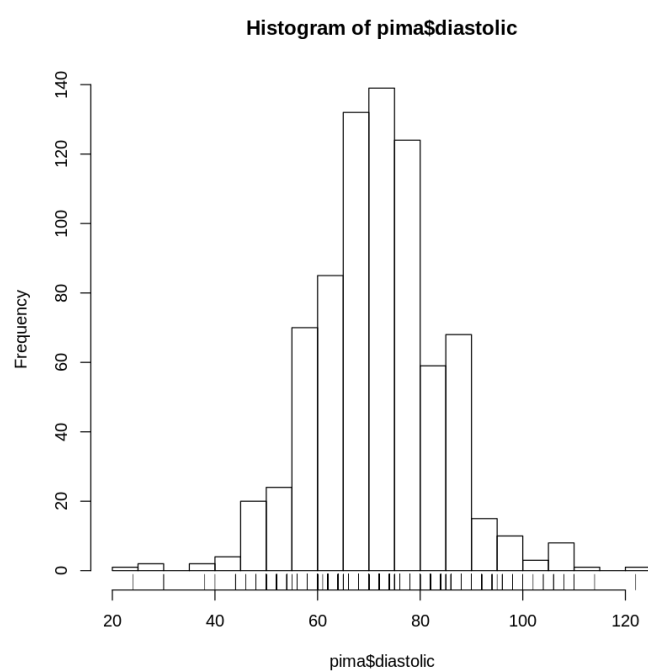
```
In [25]: zeroDiastolicIndices = which(pima$diastolic==0)
         zeroDiastolicIndices
```

```
In [26]: pima$diastolic[zeroDiastolicIndices]
```

```
In [27]: pima$diastolic[zeroDiastolicIndices] = round(mean(pima$diastolic[pima$diastolic!=0]))
```

```
In [28]: pima$diastolic[zeroDiastolicIndices]
```

```
In [29]: hist(pima$diastolic, breaks = 18)
rug(pima$diastolic)
```

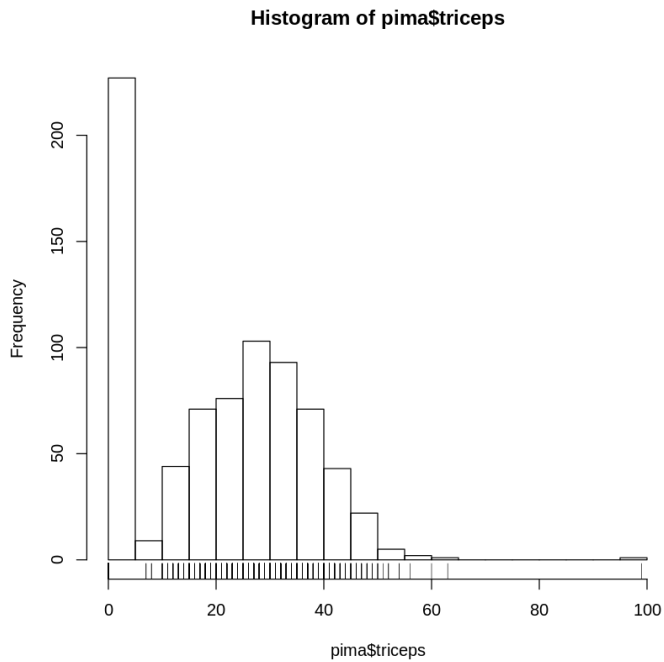


triceps

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	0.00	23.00	20.54	32.00	99.00

7/15

```
In [31]: hist(pima$triceps, breaks = 18)
rug(pima$triceps)
```



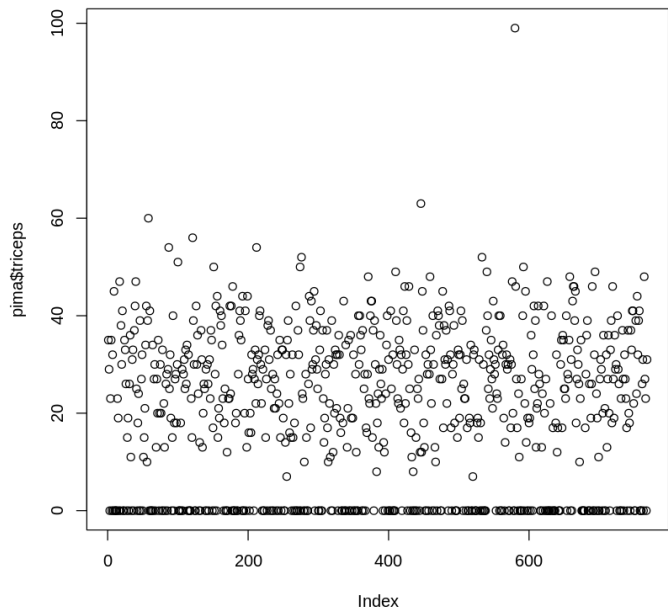
Lets look at some of the top triceps values:

```
In [32]: pima[pima$triceps>70,]
```

A data.frame: 1 × 9

	pregnant	glucose	diastolic	triceps	insulin	bmi	diabetes	age	test
	<int>	<dbl>	<dbl>	<int>	<int>	<dbl>	<dbl>	<int>	<int>
580	2	197	70	99	0	34.7	0.575	62	1

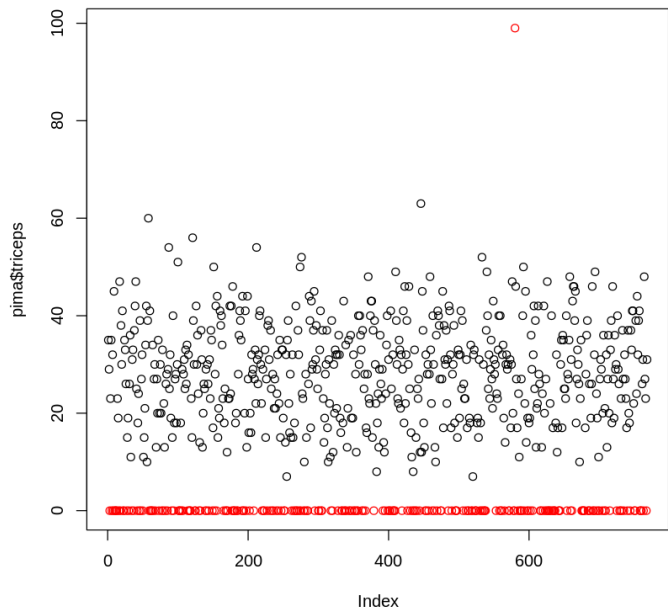
```
In [33]: plot(pima$triceps)
```



```
In [34]: table(pima$triceps>80 | pima$triceps==0)
```

FALSE	TRUE
540	228

```
In [35]: plot(pima$triceps, col=ifelse(pima$triceps>80 | pima$triceps==0, 'red', 'black'))
```



Yes. This looks good. Lets replace these zero triceps values with the mean glucose values.

```
In [36]: zeroTricepsIndices = which(pima$triceps>80 | pima$triceps==0)
zeroTricepsIndices
```

3· 6· 8· 10· 11· 12· 13· 16· 18· 22· 23· 27· 30· 34· 37· 42· 45· 47· 50· 59· 61· 62· 63· 65· 68· 73· 77· 79· 82· 85·
91· 94· 101· 102· 103· 105· 107· 114· 116· 117· 118· 124· 125· 130· 132· 139· 141· 144· 149· 152· 155· 165· 168· 169·
171· 177· 179· 180· 181· 184· 185· 191· 193· 194· 197· 202· 208· 220· 222· 223· 227· 231· 234· 236· 240· 243· 247· 251·
252· 262· 265· 267· 269· 270· 273· 275· 279· 281· 284· 285· 295· 300· 301· 304· 305· 318· 320· 328· 333· 334· 337· 338·
340· 344· 345· 348· 351· 352· 355· 356· 362· 364· 367· 379· 392· 395· 399· 401· 402· 405· 407· 408· 409· 419· 427· 431·
434· 436· 438· 440· 444· 452· 454· 457· 462· 465· 469· 474· 475· 485· 490· 495· 496· 497· 506· 510· 513· 514· 518· 519·
523· 524· 525· 530· 532· 534· 536· 537· 538· 553· 558· 560· 561· 565· 571· 572· 578· 579· 580· 584· 587· 588· 590· 593·
597· 599· 602· 605· 616· 617· 620· 623· 625· 627· 628· 629· 631· 633· 635· 636· 637· 642· 643· 644· 654· 659· 661· 675·
676· 677· 678· 679· 684· 685· 687· 691· 692· 695· 698· 700· 704· 707· 709· 715· 725· 729· 730· 732· 735· 740· 744· 750·
751· 758· 759· 760· 763· 767

```
In [37]: pima$triceps[zeroTricepsIndices]
```

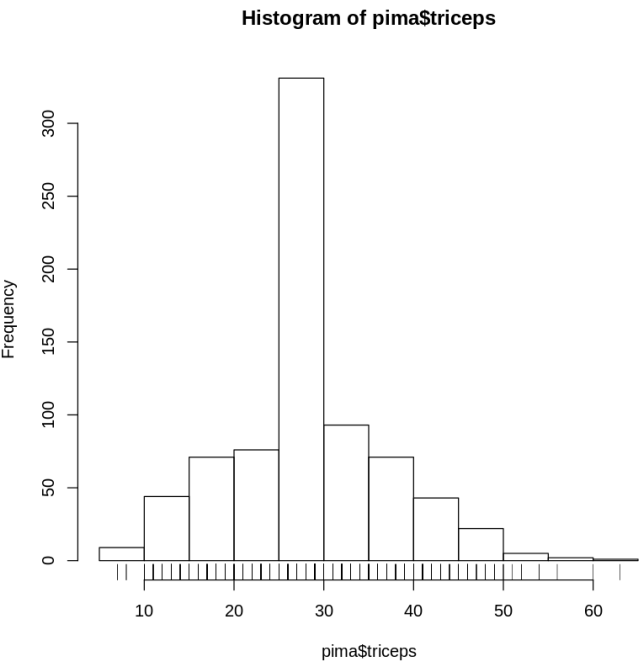
0·
0·
0·
0·
0· 99· 0· 0· 0· 0· 0· 0· 0· 0· 0· 0· 0· 0· 0· 0· 0· 0· 0·
0·
0· 0· 0· 0· 0· 0· 0

```
In [38]: pima$triceps[zeroTricepsIndices] = round(mean(pima$triceps[-zeroTricepsIndices]))
```

```
In [39]: pima$triceps[zeroTricepsIndices]
```

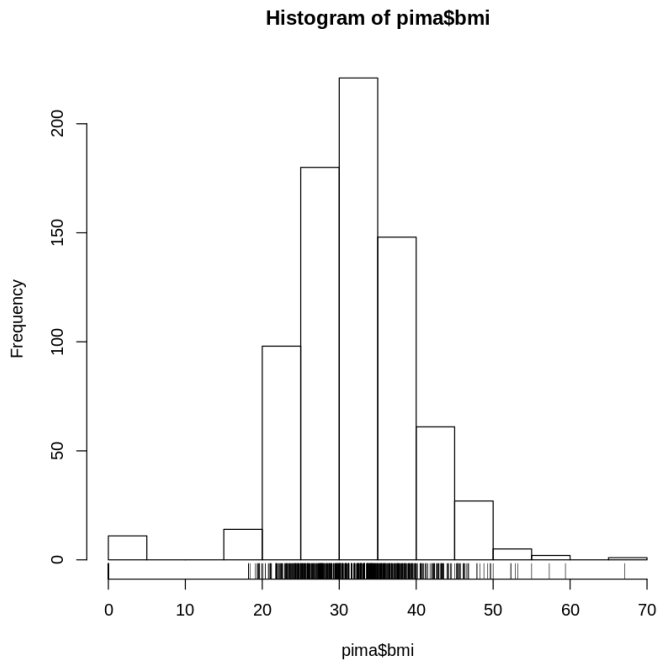
29·
29·
29·
29·
29·
29·
29· 29

```
In [40]: hist(pima$triceps, breaks = 18)
rug(pima$triceps)
```



bmi

```
In [41]: hist(pima$bmi, breaks = 18)
rug(pima$bmi)
```



```
In [42]: table(pima$bmi == 0)

FALSE TRUE
 757    11
```

- 11 women are showing zero diastolic. This is most likely an error.

```
In [43]: pima[pima$bmi==0,]
```

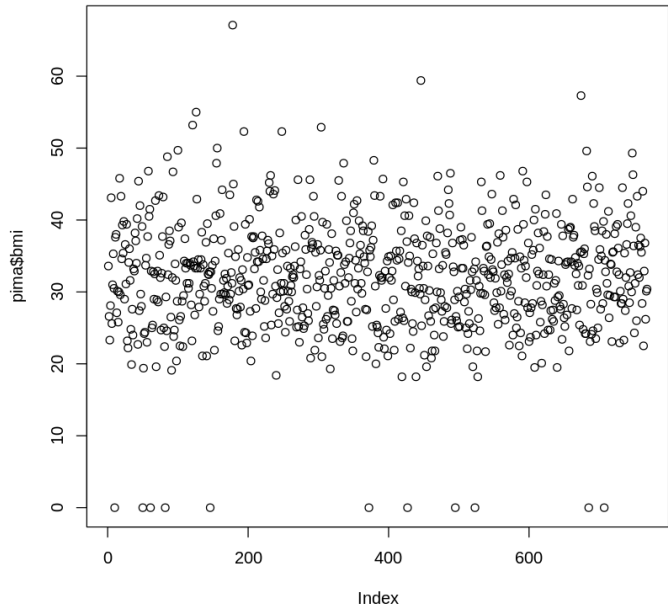
A data.frame: 11 × 9

	pregnant	glucose	diastolic	triceps	insulin	bmi	diabetes	age	test
	<int>	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<int>	<int>
10	8	125	96	29	0	0	0.232	54	1
50	7	105	72	29	0	0	0.305	24	0
61	2	84	72	29	0	0	0.304	21	0
82	2	74	72	29	0	0	0.102	22	0
146	0	102	75	23	0	0	0.572	21	0
372	0	118	64	23	89	0	1.731	21	0
427	0	94	72	29	0	0	0.256	25	0
495	3	80	72	29	0	0	0.174	22	0
523	6	114	72	29	0	0	0.189	26	0
685	5	136	82	29	0	0	0.640	69	0
707	10	115	72	29	0	0	0.261	30	1

In principle we can drop this. But lets not do that.

Lets see if we can use that data to replace the zero bmi value.

```
In [44]: plot(pima$bmi)
```



Yes. This looks good. Lets replace these zero bmi values with the mean glucose values.

```
In [45]: zeroBmiIndices = which(pima$bmi==0)
zeroBmiIndices

10 · 50 · 61 · 82 · 146 · 372 · 427 · 495 · 523 · 685 · 707

In [46]: pima$bmi[zeroBmiIndices]

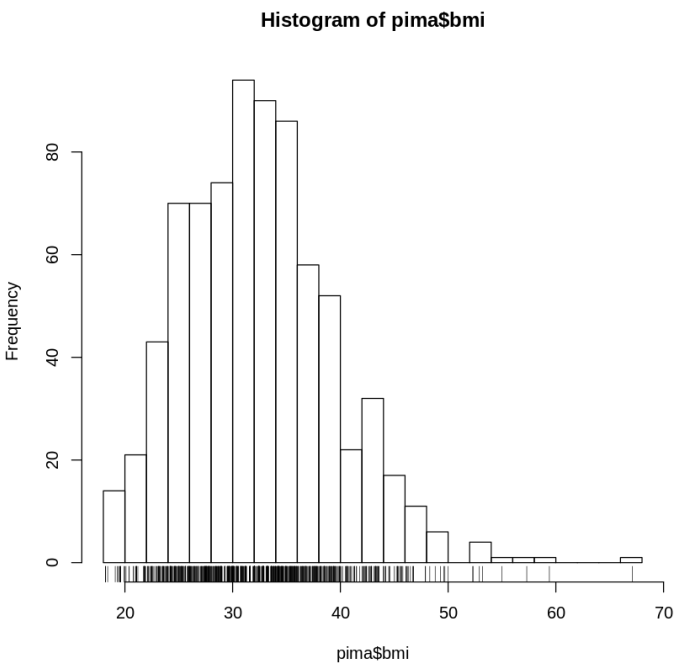
0 · 0 · 0 · 0 · 0 · 0 · 0 · 0 · 0 · 0 · 0

In [47]: pima$bmi[zeroBmiIndices] = round(mean(pima$bmi[-zeroBmiIndices]))

In [48]: pima$bmi[zeroBmiIndices]

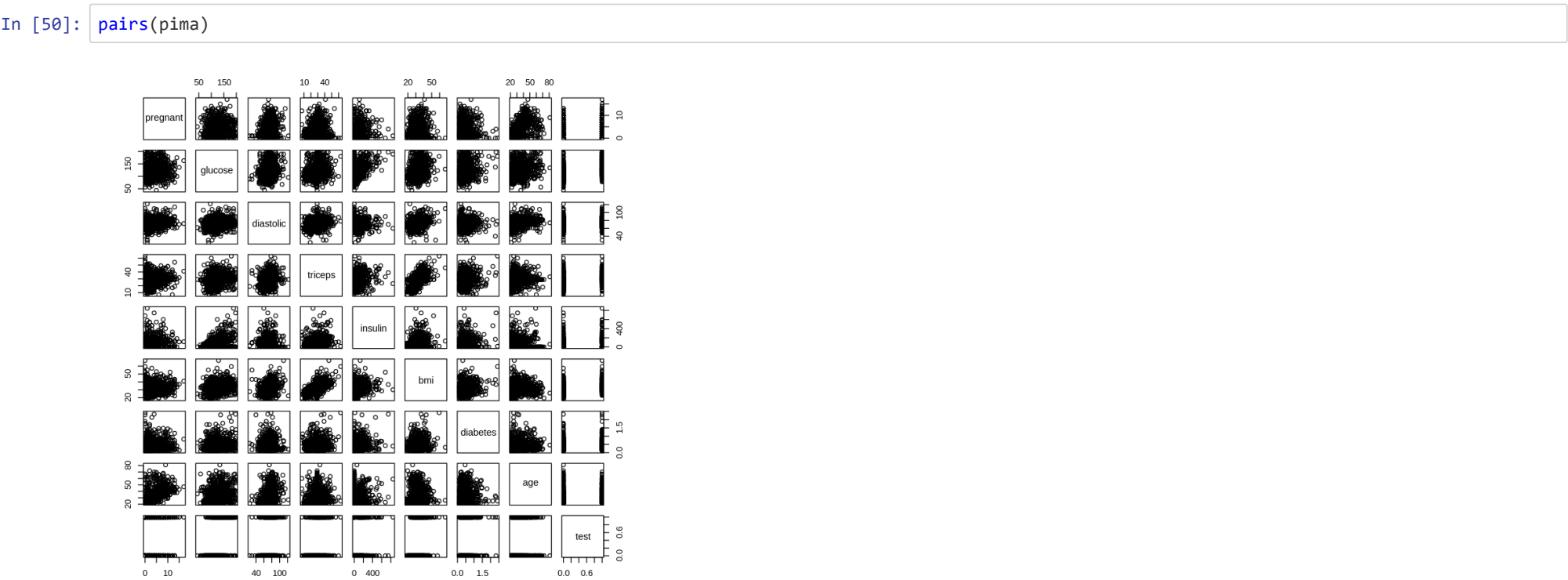
32 · 32 · 32 · 32 · 32 · 32 · 32 · 32 · 32 · 32 · 32

In [49]: hist(pima$bmi, breaks = 18)
rug(pima$bmi)
```



Looks good!

Pair Plot



Test category

```
In [51]: table(pima$test)

 0  1
500 268

In [52]: round(100*prop.table(table(pima$test)))

 0  1
65 35
```

Nearly 35% women seem to be testing positive!

1. (b) Initial GLM modelling

Our data is clean and we're ready to fit! What kind of model should we use to fit these data? Notice that the `test` variable is either 0 or 1, for whether the individual tested positive for diabetes. Because `test` is binary, we should use logistic regression (which is a kind of binomial regression).

Fit a model with `test` as the response and all the other variables as predictors. Can you tell whether this model fits the data?

```
In [53]: # Your Code Here

In [54]: pimaModel = glm(test~., data=pima, family = 'binomial')
summary(pimaModel)

Call:
glm(formula = test ~ ., family = "binomial", data = pima)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.6710  -0.7135  -0.3998   0.7233   2.4136

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -9.219944    0.820936 -11.231  < 2e-16 ***
pregnant     0.123346    0.032474   3.798 0.000146 ***
glucose      0.038813    0.003870  10.028  < 2e-16 ***
diastolic    -0.010765    0.008658  -1.243  0.213730
triceps      0.003328    0.013563   0.245  0.806180
insulin     -0.001560    0.000834  -1.871  0.061412 .
bmi          0.096741    0.018046   5.361 8.29e-08 ***
diabetes     0.927483    0.299925   3.092 0.001986 **
age          0.011753    0.009508   1.236 0.216386
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 993.48  on 767  degrees of freedom
Residual deviance: 710.20  on 759  degrees of freedom
AIC: 728.2

Number of Fisher Scoring iterations: 5
```

It doesn't seem like we are doing very well:

- The AIC is very high, means the model is far away from the true model.
- The Null and Residual deviances are rather close to each other.

1. (c) Remember Bayes

A quick analytical interlude.

Is diastolic blood pressure significant in the regression model? Do women who test positive have higher diastolic blood pressures? Explain the distinction between the two questions and discuss why the answers are only apparently contradictory.

- Diastolic blood pressure is not significant with p-value being 0.21.
- Lets see the data for diastolic blood pressure based on test:

```
In [57]: # Your Code Here

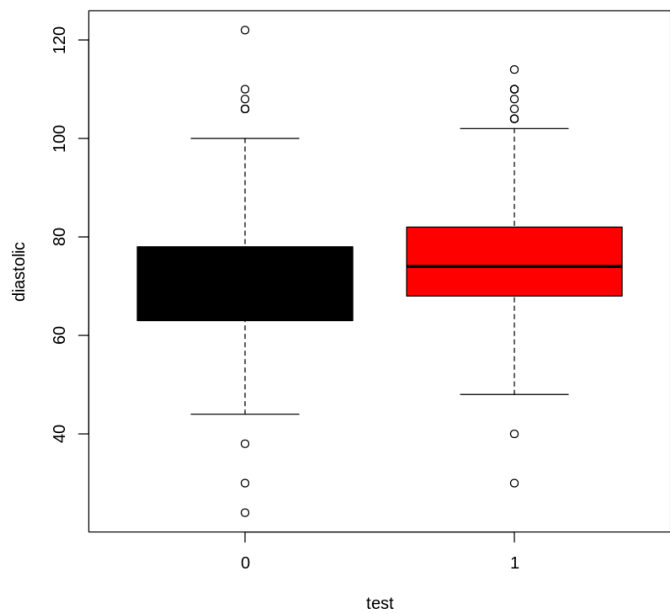
In [58]: # For women who test positive
summary(pima$diastolic[pima$test ==1])

  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 30.00  68.00   74.00   75.12  82.00  114.00

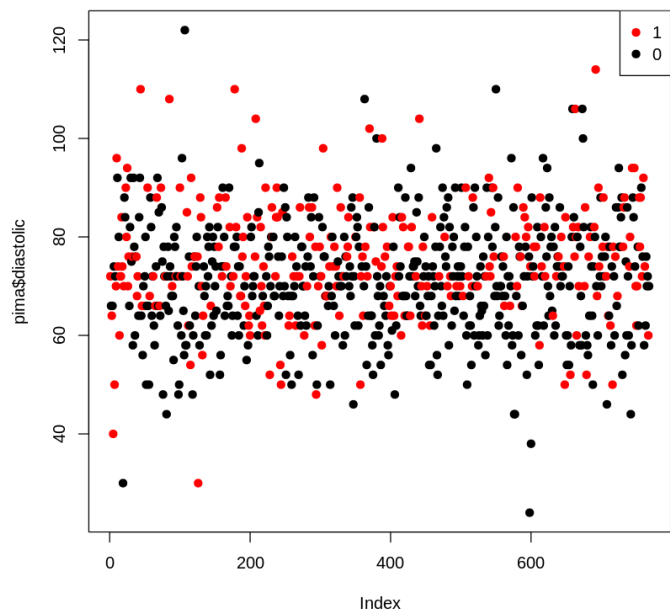
In [59]: # For women who test negative
summary(pima$diastolic[pima$test ==0])

  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 24.00  63.50   72.00   70.92  78.00  122.00
```

```
In [60]: boxplot(diastolic~test, data = pima, col= c('black', 'red'), names=c(0, 1))
```



```
In [61]: plot(pima$diastolic, col = ifelse(pima$test ==1, 'red', 'black'), pch = 19)
legend('topright', legend=c(1, 0), col = c('red', 'black'), pch =19)
```



- Clearly the two groups have very similar distributions.
- The first question talks about the significance of diastolic pressure in predicting the test result, while the second question talks about the difference between the two groups. They are related. Since diastolic pressure is similar for passing and failing groups, diastolic pressure is not expected to have high predictive power.

1. (d) GLM Interpretation

We’ve seen so many regression summaries up to this point, how is this one different from all the others? Well, to really understand any model, it can be helpful to loop back and plug the fitted results back into the model’s mathematical form.

Explicitly write out the equation for the binomial regression model that you fit in (b). Then, in words, explain how a 1 unit change of `glucose` affects `test`, assuming all other predictors are held constant.

This summary is different in couple of ways:

- Here evaluation metrics are different: deviances, AIC etc are showing instead of goodness of fit.
- Here results is obtained through an iterative process, and the corresponding count is shown.
- Here the coefficients are on linear scale, have different interpretation than the linear models.

```
In [62]: # Your Code Here
```

```
In [63]: pimaModel

Call:  glm(formula = test ~ ., family = "binomial", data = pima)

Coefficients:
(Intercept)    pregnant    glucose    diastolic    triceps    insulin
-9.219944    0.123346    0.038813   -0.010765    0.003328   -0.001560
      bmi    diabetes      age
 0.096741  0.927483  0.011753

Degrees of Freedom: 767 Total (i.e. Null); 759 Residual
Null Deviance: 993.5
Residual Deviance: 710.2      AIC: 728.2
```

Equation for the binomail regression model:

$$g(\frac{p}{1-p}) = \eta_i = \beta_0 + \beta_1 x_{pregnant,i} + \beta_2 x_{glucose,i} + \beta_3 x_{triceps,i} + \beta_4 x_{insulin,i} + \beta_5 x_{bmi,i} + \beta_6 x_{diabetes,i} + \beta_7 x_{age,i}$$

```
In [64]: exp(pimaModel$coefficients['glucose'])  
  
glucose: 1.03957583292414
```

One unit change in glucose increases the odds of failing the test by a (multiplicative) factor of $e^{\beta_1} = e^{0.038} = 1.04$, assuming all other predictors are held constant.

1. (e) GLM Prediction

One of the downsides of Logistic Regression is that there isn't an easy way of evaluating the goodness of fit of the model without predicting on new data. But, if we have more data to test with, then there are many methods of evaluation to use. One of the best tools are confusion matrices, which (despite the name) are actually not that hard to understand.

A confusion matrix compares the predicted outcomes of a Logistic Regression Model (or any classification model) with the actual classifications. For binary classification, it is a 2×2 matrix where the rows are the models' predicted outcome and the columns are the actual classifications. An example is displayed below.

	True	False
1	103	37
0	55	64

In the example, we know the following information:

- The [1,1] cell is the number of datapoints that were correctly predicted to be 1. The value (103) is the number of True Positives (TP).
- The [2,2] cell is the number of datapoints that were correctly predicted to be 0. The value is the number of True Negatives (TN).
- The [1, 2] cell is the number of datapoints that were predicted to be 1 but where actually 0. This is the number of False Positives (FP), also called Type I error. In the context of our diabetes dataset, this would mean our model predicted that the person would have diabetes, but they actually did not.
- The [2, 1] cell is the number of datapoints that were predicted to be 0 but where actually 1. This is the number of False Negatives (FN), also called Type 2 error. In the context of our diabetes dataset, this would mean our model predicted that the person would not have diabetes, but they actually did have diabetes.

Use your model to predict the outcomes of the test set. Then construct a confusion matrix for these predictions and display the results.

```
In [65]: # Your Code Here  
  
In [66]: table(as.integer(pimaModel$fitted.values > 0.5))  
  
  0  1  
560 208  
  
In [67]: pimaConfMat = table(as.integer(pimaModel$fitted.values > 0.5), pima$test==1)  
pimaConfMat  
  
      FALSE TRUE  
0      444  116  
1       56  152  
  
In [68]: table(pima$test)  
  
  0  1  
500 268
```

1. (f) Evaluation Statistics

Using the four values from the confusion matrix, we can construct evaluation statistics to get a numerical approximation for our model's performance. Spend some time researching accuracy, precision, recall and F score.

Calculate these values for your model's predictions on the test set. Clearly display your results. How well do you think your model fits the data?

```
In [69]: # Your Code Here  
  
In [70]: pima.accuracy = sum(diag(pimaConfMat))/sum(pimaConfMat)  
pima.accuracy  
  
0.776041666666667  
  
In [71]: pima.recall = pimaConfMat[2,2]/sum(pimaConfMat[,2])  
pima.recall  
  
0.567164179104478  
  
In [72]: pima.precision = pimaConfMat[2,2]/sum(pimaConfMat[2,])  
pima.precision  
  
0.730769230769231  
  
In [73]: pima.f_score = 2 /(1/pima.precision + 1/pima.recall)  
pima.f_score  
  
0.638655462184874  
  
In [74]: 2 *(pima.precision * pima.recall)/(pima.precision + pima.recall)  
  
0.638655462184874
```

- Model has a low recall (56.7%), so many positive cases are not being detected.
- Model has a moderate precision (73.1%), so among the case identified as positive about 27% are not positive.
- So model needs to be improved significantly before being useful in any real application.

1. (g) Understanding Evaluation Statistics

Answer the following questions in the markdown cell below.

1. Give an example scenario for when accuracy would be a misleading evaluation statistic.

2. Confusion matrices can also be used for non-binary classification problems. Describe what a confusion matrix would look like for a response with 3 levels.

3. You'll have to take our word on the fact (or spend some time researching) that Type I error and Type II error are inversely related. That is, if a model is very good at detecting false positives, then it will be bad at detecting false negatives. In the case of our diabetes dataset, would you prefer a model that overestimates the Type 1 error or overestimates the Type II error. Justify your answer.

1. Accuracy will be a misleading evaluation statistic when there is high class imbalance. Say the target class is <1% of the total population.

1. For a 3 level problem, we'll have a 3 - way confusion matrix:

```
In [75]: table(sample(1:3, 100, replace = TRUE), sample(1:3, 100, replace = TRUE))

      1  2  3
1  8  9 11
2 16 12 10
3 10 12 12

In [76]: table('true'=sample(1:3, 100, replace = TRUE), 'pred'=sample(1:3, 100, replace = TRUE))

      pred
true  1  2  3
1    7 12  9
2   15 11 12
3   14 10 10
```

1. Both errors need to be avoided.
- Type 1 error could lead to physical, mental and finacial stress.

Type II error could lead the disease going undiagnosed for long period. Thus delaying the physical, mental and finacial stress.

Thus, Type 1 error would be preferred, that is we have higher false positives.

- Secondary tests can be used to verify the findings.

1. (h) Ethical Issues in Data Collection

Read Maya Iskandarani's [piece](https://researchblog.duke.edu/2016/10/24/diabetes-and-privacy-meet-big-data/) (https://researchblog.duke.edu/2016/10/24/diabetes-and-privacy-meet-big-data/) on consent and privacy concerns raised by this dataset. Summarize those concerns here.

Maya raises the point about consent and privacy concerns. When collecting the data for medical studies, the research teams need to clearly inform the patients about what data will be collected and for what purposes it will be used. In most cases, maintaining privacy and hiding personally identifiable information is mandatory. Researchers need to take concent from the patient, and only then they can use the data for the consented purpose and duration. However, the PIDD data used in this exercise is being used today in ways that were unthinkable for researchers when the data was being collected. This data is useful in developing new algorithms, but did the patient give consent for this? Thus the study highlights the challenge of informed concent for future uses, and making such sensitive data publically available for research. This might have real implications for these individuals and their community, about policy decisions at public and private institutions.

Problem 2: Practicing those Math skills

One of the conditions of GLMs is that the "random component" of the data needs to come from the Exponential Family of Distributions. But how do we know if a distribution is in the Exponential Family? Well, we could look it up. Or we could be proper mathematicians and check the answer ourselves! Let's flex those math muscles.

2. (a) But it's in the name...

Show that $Y \sim exponential(\lambda)$, where λ is known, is a member of the exponential family.

- We know that: $Y(y; \lambda) = \lambda e^{-\lambda y}; y > 0$

We can rewrite it as: $Y(y; \lambda) = e^{-\lambda y + \log(\lambda)}; y > 0$

Which is clearly the form of exponential family distributions.

2. (b) Why can't plants do math? Because it gives them square roots!

Let $Y_i \sim exponential(\lambda)$ where $i \in \{1, \dots, n\}$. Then $Z = \sum_{i=1}^n Y_i \sim Gamma(n, \lambda)$. Show that Z is also a member of the exponential family.

$$Z = \sum_{i=1}^n Y_i \sim Gamma(n, \lambda) = \frac{\lambda^n}{\Gamma(n)} z^{n-1} e^{-\lambda z} = e^{\log\left(\frac{\lambda^n}{\Gamma(n)} z^{n-1}\right) - \lambda z} = e^{-\lambda z + n \log \lambda - \log \Gamma(n) + (n-1) \log z}$$

```
In [ ]:
```